Practical Assignment III

Introduction to Statistical Methods

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1 Pre-assignment Reminders

- Prior to attempting this homework, I strongly recommend that you have completed all other work through module 8, including:
  - Having watched all lectures thoroughly and taken notes (optionally with the guided notes)
  - Reading all assigned sections of the book, and completing practice examples as helpful and useful
  - Completing relevant lecture check-ins, and using answer key for review
  - Completing all quizzes, and reviewing correct answers at the end of the week
- This practical assignment will be graded for accuracy, please give yourself enough time to complete your best work
  - You may use your notes, book, and lectures to aid you in completing this
  - Where possible, include more detail to ensure you fully explain your rationale for a question
  - It's worth noting that I don't place grading emphasis on grammar or spelling, but please make sure your ideas are as clear as possible.
- This practical assignment is cumulative to the modules and practical assignment(s) prior, please review your previous work and notes to help you accomplish this work
- All syllabus and university policies on academic integrity, plagiarism, and other forms of misconduct apply to this assignment. Please review them if you are unfamiliar

2 Context

The following information will be useful for answer some questions on the homework, please ensure you read and understand the following context

You are working for a healthcare organization that is partnering with a local university to better understand how undergraduate students get and feel about healthcare-focused internships. In specific, these are nursing and pre-med students gaining experience in their respective fields.

The healthcare org is interested in generalizing conclusions from this study to all healthcare internship students in the US. They notice that there are currently 80 students participating in internships in their specific hospital right now, and decide to gather data on a subset these students, to draw conclusions from. The first 10 student interns who clock in one day are pulled to the side and asked to answer some questions.

First, student interns are asked these following characteristics about themselves:

1. What their undergraduate cumulative GPA is (from 0.00 to 4.00)
2. What their current age is (from 18 to 40)
3. Whether they have a current internship (yes/no)

4. Whether they are an undergraduate student (yes/no)
5. Whether they are a first-generation college student or not (yes/no)
6. How many jobs they currently work (only whole number counts from 0 to 3, excluding the internship)

Next, the student interns are asked questions about their opinions:

1. How likely are you to recommend getting an internship to other students (Not likely, somewhat, very likely)
2. How likely are you to continue in this line of work when you graduate (Not likely, somewhat, very likely)

3 Instructions

> **Important**
>
> Please - provide as much detail as you can to adequately address each question; I'd prefer more information rather than less. Remember to follow all expectations for creating original work, as described in the syllabus.

Answer all of the following questions in full, using appropriate vocabulary from past modules:

3.1 Sampling and Design

1. Identify the population of interest and the sample used in this study - is this a census? Explain your thinking (2 pts)
2. Address whether sampling was done with or without replacement, and how that impacts independence - Explain your thinking (2 pts)
3. Do you anticipate that the results will generalize from the sample to the population? - Explain your thinking and ensure you address the connection between the sample statistics and population parameters; do NOT only address size of sample (2 pts)

3.2 Descriptive Statistics

1. Identify each of the variables collected as data with the scale of measurement most appropriate: nominal, ordinal, interval/ratio. Provide brief rationale for each decision (2 pts)
2. For any interval/ratio variable, address whether it is discrete or continuous. Provide brief rationale for each decision (1 pt)

3. Name at least two variables that would be well-visualized with a frequency histogram, and explain why (1 pt)
4. Name at least two variables that would be well-visualized with a bar graph and explain why (1 pt)
5. Identify two variable for which a mean and standard deviation could NOT be calculated (1 pt)
6. Write out a fake frequency table for one of the nominal variables. Use any numbers you'd like, just make sure your table contains a column for simple frequency AND relative frequency (1 pt)

3.3 Distributions

1. True or false (and explain why): A discrete variable can be represented with an exponential distribution (1 pt)
2. In the notation used for variable distributions, we use X (uppercase) and x (lowercase) - please choose one *discrete* variable from above and write out what X and x are (1 pt)
3. Write a fake probability distribution function table for this discrete variable. Use any numbers you'd like, but ensure your probabilities sum to one and that the table follows the format from lecture (1 pt)
4. In the notation used for variable distributions, we use X (uppercase) and x (lowercase) - please choose one *continuous* variable from above and write out what X and x are (1 pt)
5. Write a probability density function in notation of a *uniform* variable, using the information provided about the variable (1 pt)

3.4 Reflection

1. Which two of the previous questions were the most challenging for you, and why? (2 pts)

I would encourage you to review and study hard on the areas that were most challenging for you, in preparation for the midterm exam!